



INFORMATION
TECHNOLOGY
UNIVERSITY

EagleEye

Intelligent Edge AI Surveillance System

On-device AI · Cloud-connected · Watch your space from anywhere

SparkUp / ORIC
logo
assets/sparkup_logo.png

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1 THE PROBLEM

Security cameras force a bad trade-off:

Cloud cams stream your footage to a company's servers — privacy risk + monthly fees.

LAN / DIY cams work only on the same Wi-Fi — typed IP, a PC left on, risky port-forwarding.

Most **record everything** — real events buried in hours of empty video.

Goal: see real intrusions — anywhere, privately, no subscription.

2 OUR SOLUTION

EagleEye runs a neural network **on the camera itself**. A low-cost ESP32-CAM decides on-device whether a human is present — images are analysed locally, never streamed to the cloud.

On an intruder it pushes a **photo alert to your phone**; you can open **live video** and **pan the camera** — from any network (Wi-Fi or 4G/LTE).

The camera **reaches OUT** to the cloud — no static IP, no port-forwarding, no on-site PC. Encrypted end-to-end; runs on free tiers.



EagleEye node — ESP32-CAM in a 3D-printed mount · swap for a real photo (assets/prototype_photo.jpg)

3 KEY FEATURES

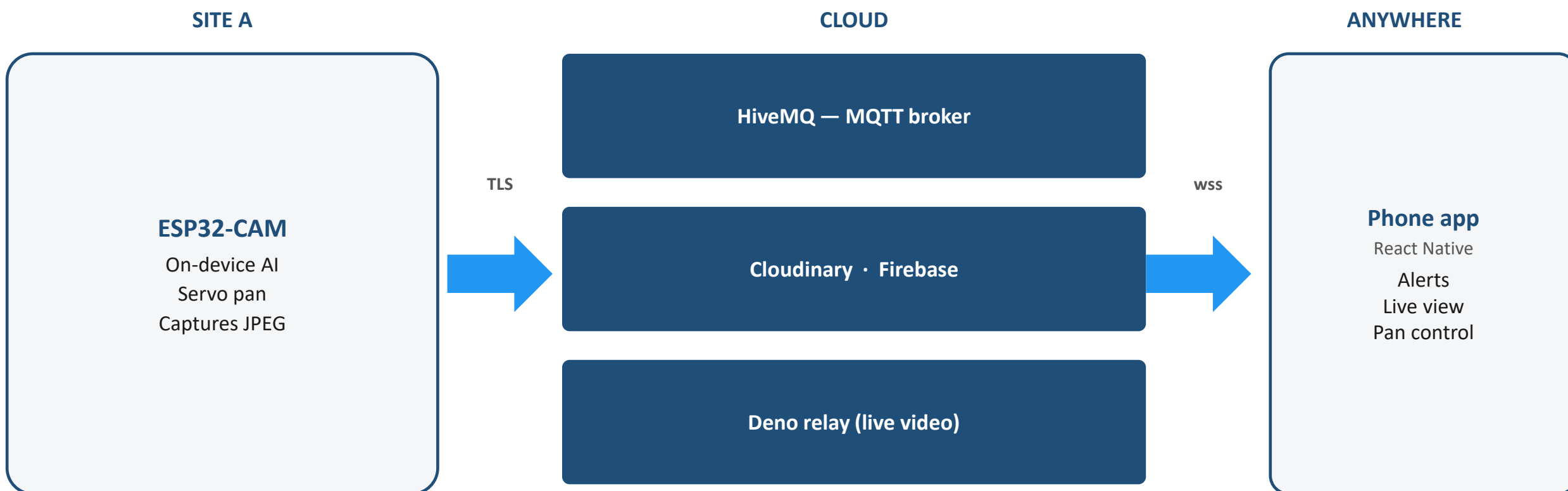
- ✓ **On-device human detection** — TinyML; images never leave the device
- ✓ **Instant intruder alerts** — captured photo pushed to your phone
- ✓ **On-demand remote live video** — stream only when you ask
- ✓ **App-controlled camera pan** — servo aim from the app
- ✓ **Arm / disarm + true online status** — real online/offline, not guesswork
- ✓ **Encrypted & private** — TLS everywhere, per-device credentials
- ✓ **~\$10 hardware · \$0/month cloud** — entire stack on free tiers

IMPACT & SUSTAINABILITY

Affordable AI security — ~\$10 vs \$200+ systems democratizes safety for low-income homes (SDG 9 · 11).

Privacy by design — only confirmed threats ever leave the device (SDG 16).

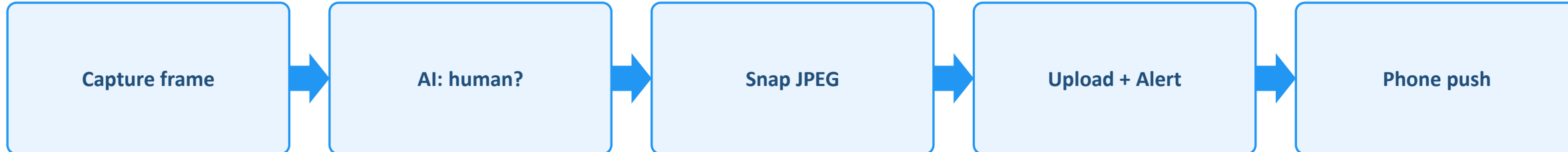
4 SYSTEM ARCHITECTURE



The camera reaches **OUT** to the cloud; the phone reaches the same cloud — they meet in the middle.

Works on any network · no static IP · no port-forwarding · no on-site PC · end-to-end TLS

5 HOW IT WORKS



On request → **live video via cloud relay** | No human → keep watching (only confirmed humans trigger an alert).

6 RESULTS & TECH STACK

90.8%

detection accuracy (test; 87.5% human recall)

<1 s

on-device inference (~872 ms, ESP32-CAM)

<3 s

detection → phone alert (end-to-end)

18

model versions (v1.0 → v7.17)

~\$10

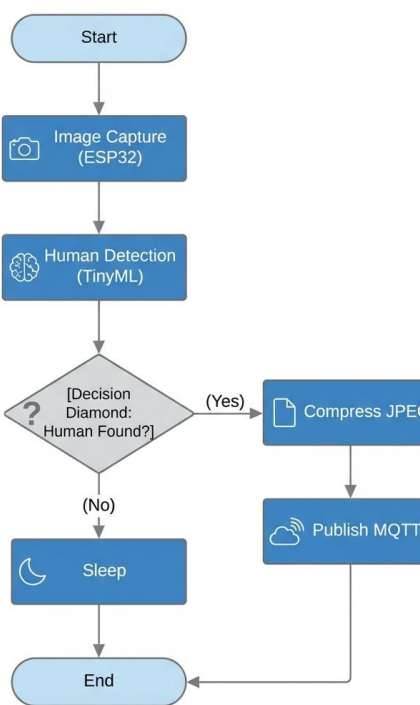
device vs \$200+ · \$0/month cloud

100%

private — frames stay on device

Tech stack: ESP32-CAM · Edge Impulse (TinyML) · MQTT / HiveMQ · Cloudinary · Firebase · Deno relay · React Native + Expo · TLS / WebSocket

Why it's novel: Edge AI on a microcontroller (privacy + zero inference cost) + an outbound-only architecture that works on any network — the pattern commercial IoT (AWS IoT, Ring/Nest) uses, rebuilt at \$0/month.



Detection pipeline (on-device)

App screenshot

dashboard / live view
assets/app_screenshot.png

Real detection

one clean capture
assets/detection_sample.jpg



GitHub repo

Demo QR
add URL

Demo video